

ShalltyUtils v1.0

A utility plugin focused on enhancing Timeline to improve animator's workflows by incorporating handy extra features.

If you use this plugin and want to share the scene with other users, they don't necessarily need this plugin, **UNLESS** the ❖ **New Timeline Interpolables** were used in the scene.

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HOW TO INSTALL

1. Download and install all the **Requirements**:
 - [BepInEx 5](#)
 - [IllusionModdingAPI](#)
 - [HSPlugins](#)
 2. Download [ShalltyUtils](#) plugin according to your game.
 3. Extract the **.DLL** file inside the **BepInEx\plugins** folder in your game directory.
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PLUGIN FEATURES

❖ ShalltyUtils Timeline Panel

● Motion Paths:

This allows you to visualize the position of the Keyframes as paths. It only works with 'GuideObject Position' Interpolables. Provides different modes to determine which keyframes to include in the motion path:

- **Selected Keyframes:**
Displays the motion path of the selected keyframes.
- **Selected Interpolable:**
Displays the motion paths of all the keyframes of the first selected Interpolable.
- **Selected Interpolable (Range):**
Displays the motion path of all the keyframes of the first selected Interpolable within a specified range.
- **Hover** over the motion path **Keyframes** to see information about them and seek to a Keyframe time by **right-clicking** on it.
- Navigate through the **Timeline** by **scrolling** while holding the **Ctrl** key.
- **Hold Ctrl** and move the **GuideObject** to edit the Keyframe position.
- If you **move** the **GuideObject** while **holding Ctrl** and there is not a Keyframe at the current time, then a new Keyframe will be **added**.

● Animation Baking

Bake the animation of the current **selected Character** into **Timeline**, it records the animation at the designated **Keyframes per second**. You can toggle which bones group to bake in the **BAKE INTERPOLABLES** panel. *(Undo/Redo available)*

- **Anim Speed:** Speed of the animation.
 - **Loop Count:** How many times the animation will be looped.
 - **FPS:** How many Keyframes per second will be created.
 - **MMDD Length:** Length of the VMD animation in seconds.
(ONLY FOR MMDD. Keep it in -1 if NOT using it)
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● DynamicBones Baking

Bake the DynamicBones of the **selected object** into Timeline, it records the DynamicBones at the designated Keyframes per second. You can toggle which DynamicBones bake in the **Advanced KKPE: Dynamic Bones** panel, only the **Enabled** DynamicBones will be baked. *(Undo/Redo available)*

- **Seconds:** How many Timeline seconds will be baked.
 - **Wait Seconds:** How many seconds to wait between each Keyframe bake.
 - **FPS:** How many Keyframes per second will be created.
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● Routes Baking

Bake the selected studio Route into Timeline, it records the position of each child under Child Access Point at the designated Keyframes per second.
(Undo/Redo available)

FPS: How many Keyframes per second will be created.

● Split Interpolable(s) Axis

Split the X/Y/Z axis of the **selected Interpolable(s)** for more control.

Compatible Interpolables:

- Selected GuideObject Pos.
 - Selected GuideObject Rot.
 - Bone Position (KKPE)
 - Bone Rotation (KKPE).
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● Bake X/Y/Z Interpolables

Bake all the **X/Y/Z Interpolable(s)** into **ConstraintFolders**, so that **ShalltyUtils** plugin is not a dependency when **sharing** the scene to other users.

● Save Interpolables File

Save only the currently **selected interpolable(s)** into a **.XML** file.

● Load Interpolables File

Load a Interpolables **.XML** file (also named **single files**), into the **current Time** at **Timeline**. If the interpolable already exists it **adds** the keyframes to the Interpolable (Unlike the Timeline Single Files load function that doesn't load the file).

- **Cleanup Redundant Keyframes**

Remove all the **redundant** Keyframes that won't affect the animation. It also removes **bugged** Keyframes. (When multiple Keyframes of the same Interpolable have the exact same time.) *(Undo/Redo available)*

- **Toggle Performance Mode**

Turn **ON/OFF** the **Timeline Performance Mode**, which can increase performance, it will also display the total amount of Keyframes in the current scene.

- **Mesh Sequencer**

Open the Mesh Sequencer UI: It creates a list of the meshes of the selected object to create a sequence in **Timeline**.

Use it to create animations with meshes imported with OBJ Import plugin.

- **Change Timeline UI Colors**

Change the colors of the **Timeline UI**.

❖ ShalltyUtils Keyframe Panel

● Edit Value

Here you can directly modify the **Value** of the **selected Keyframe(s)**.
(*Undo/Redo available*)

- **Edit Value (type) field:** Here you can input the new Value, it must match the Keyframe value type.
 - **Supports:** Vector2, Vector3, Vector4, Quaternion, Color, Boolean, Integer, Float.
 - **Get Current Value:** Get the **current** Value of the **first** selected Keyframe (Like the Use Current Value button).
 - **Get Keyframe Value:** Get the Value of the **first** selected Keyframe.
 - **Get Default Value:** Get the default Value of the current type.
 - **Set:** Set the **Edit Value** to the selected Keyframe(s).
 - **+** : Adds the **Edit Value** to the selected Keyframe(s).
 - **-** : Subtract the **Edit Value** from the selected Keyframe(s).
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● Randomize

Here you can **Randomize** the **Value** or **Time** of the selected Keyframe(s). The randomness threshold can be adjusted. (*Undo/Redo available*)

● Curve Presets

You can **Save** and **Load** your own Curve Presets, they will be saved in "*BepInEx\plugins\ShalltyUtils\Curve Presets*" folder, so you can share the curves.

● Graph Editor

The **Graph Editor** it's like the old curve editor of **Timeline** but with more controls.

- **Window Mode:** Press to dock/undock the Graph Editor Window.
 - **Reset View:** Press to Reset the Graph Editor zoom and position.
 - **Graph Editor Controls:**
 - **[Scroll Wheel]** to Zoom In/Out.
 - **[Scroll Wheel + Left Shift]** to Zoom In/Out Horizontally.
 - **[Scroll Wheel + Left Alt]** to Increase/Decrease Grid cell size.
 - **[Scroll Wheel + Left Ctrl]** Increase/Decrease Window size in **Window Mode**.
 - ❖ **Curve controls:**
 - ❖ **[Middle Click]** on the graph editor to **Add** a **curve keyframe**.
 - ❖ **[Scroll Wheel + Left Ctrl]** on a **curve keyframe** to change **Left Tangent**.
 - ❖ **[Scroll Wheel + Left Alt]** on a **curve keyframe** to change **Right Tangent**.
 - ❖ **[Scroll Wheel]** on a **curve keyframe** to change the orientation of **both tangents**.
 - ❖ **[Left Ctrl + Middle Click]** on a **curve keyframe** to **remove** it.
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● Edit Keyframes Times Mode:

Turn ON/OFF the **Edit Keyframes Times Mode**, where you can edit the **Times** of the selected Keyframes with a curve in the **Graph Editor**. You have to select at least 3 Keyframes. *(Undo/Redo available)*

❖ ShalItyUtils NodeConstraints Panel

- **Create FolderConstraints to IK Bones**

Create **FolderConstraints** for all IK Guide Objects of the currently selected **Character**.

- **Create FolderConstraints to Misc Bones**

Create **FolderConstraints** to the bones: (**Bust, Siri, Spine, Waist, Thigh**)

- **Create Parent Folder**

Create a **Parent folder** to the **selected object(s)** in the **Workspace**. Useful to add animation layers when working with **Timeline** and **FolderConstraints**.

- **Create FolderConstraint to:**

Create a **FolderConstraint** to the currently selected bone in **Nodes Constraint UI**.

- **Child type:**

- **Folder:** The child of the **FolderConstraint** will be a **Folder**, so it **cannot** be scaled.
- **Sphere:** The child of the **FolderConstraint** will be a **Sphere**, so it can be scaled.
Position/Rotation/Scale: The links that the new **FolderConstraint** will have.
(Scale is only available for child type Sphere)
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- **Reversed FolderConstraint:**

Reverse the **FolderConstraint** so instead of the **Bone** following the **Folder**, will be the **Folder** following the **Bone**. Use this to attach multiple objects to bones.

❖ ShalIyUtlis Misc Features

- ❖ Undo/Redo for Timeline.
- ❖ Display interpolables names when hovering.
- ❖ Automatically select linked GuideObject when selecting an Interpolable.
- ❖ Keep Timeline Interpolable(s) after creating a FolderConstraint.
- ❖ Reset Timeline playback time after loading or importing a new scene.
- ❖ Rename all the currently selected Interpolables (Shortcut, Default 'F2').
- ❖ Create a GuideObject after selecting a bone in the KKPE Advanced Window.
- ❖ Added a button in the Workspace panel to Collapse/Expand all TreeNodes.
- ❖ Keybind to add Keyframes at current time to the selected Interpolable(s).
- ❖ Double click into a Keyframe to select all the Keyframes with the same value in the Interpolable (when Performance Mode ON).
- ❖ You can now select multiple GuideObjects.

❖ New Timeline Interpolables:

 **IMPORTANT:**

If you use these **Interpolables** in your scene and want to **share** the scene, the other user will need **ShalIyUtlis** plugin.

The only exception being the GO Position and Rotation XYZ Interpolables, that you can bake to remove the dependency.

- ❖ GO Position and Rotation X, Y, Z (*THESE CAN BE BAKED*).
- ❖ KKPE Bones Position and Rotation X, Y, Z.
- ❖ Character FK & IK [Enable/Disable].
- ❖ NodesConstraints Position, Rotation and Scale offset.

